Benjamin Schaffer

About the EWB Local Project…

The UMass Amherst Engineers Without Borders (EWB) Local Project is partnered with *Nuestras Raíces*, a nonprofit organization supporting Holyoke's vibrant Puerto Rican and Latinx agricultural community. Our project's primary goal is to design and implement a biogas heating system to sustainably warm one of the organization's greenhouses, thereby extending the growing season while reducing reliance on fossil fuels. Our engineering team is conducting feasibility studies, system modeling, and prototype testing through weekly club meetings to create a reliable, low-cost biogas unit that converts organic farm waste into heat energy. We collaborate closely with experienced faculty, whose work is backed by proven, data-driven results. Our anticipated outcome is a self-sustaining renewable energy system that supports food security, reduces carbon emissions, and serves as an educational tool for sustainable farming practices within the Holyoke Massachusetts community. In addition, the Local project also aims to repair structural weather damages to another greenhouse at *Nuestras Raíces.*

History of Nuestras Raíces…

Nuestras Raíces is a Spanish term that translates to "our roots." *Nuestras Raíces* was founded in South Holyoke in 1992 by a group of migrating farmers from Puerto Rico who found themselves in a city with little opportunity. They transformed an abandoned plot of land into the first community garden of its type, ridding it of garbage and criminal activity. In response, the surrounding population followed suit and established many more gardens, creating an intricately tight-knit *Nuestras Raíces* network. These actions led to an improvement in the quality of life for many people by granting them means to grow and sell their crops. To this day, *Nuestras Raíces* remains deeply rooted in the Latinx and Caribbean cultural traditions of its members. Our project aims to nourish this grassroots network by providing cutting edge technology in an agricultural application that fosters both cultural and economic resilience.

Projected Impacts…

Our biogas heating solution will drive a handful of positive impacts. Successful deployment of our project will reduce heating costs, lower greenhouse gas emissions, and improve year-round crop yields for the local farmers. Beyond the metrics, our project has a deeper, human-centered mission: empowering a community through hands-on involvement, and creating inspiring opportunities for youth education in renewable energy and sustainability. In the long term, our model can be replicated at other community farms across Western Massachusetts, and possibly beyond, thus fostering environmental resilience and food security especially in times of dire uncertainty.

What sets us apart…

Our mission stands apart in both vision and execution. This project fuses the principles of renewable energy with the ethos of solidarity engineering, emphasizing collaboration, community empowerment, and a rejection of the conventional client-service dynamic. We are pioneering a rare integration of biogas technology into a small-scale greenhouse, a first for New England's urban agricultural landscape. Our approach prioritizes cultural inclusivity, hands-on learning, and adaptive design, resulting in a replicable model for sustainable energy that bridges traditional farming wisdom with modern engineering innovation.

Looking Ahead…

To ensure long-term impact, the system will be designed for easy maintenance and local repair, implementing only affordable and locally available materials. The EWB Local Project at UMass aims to provide comprehensive documentation, training, and workshops for *Nuestras Raíces* staff and farmers. By building local capacity and ownership, the project will remain functional beyond student involvement. Continued collaboration between EWB and *Nuestras Raíces* will support health monitoring, system optimization, and even potential expansion to other community sites.